



الشركة العمانية للمصافي والبتروكيماويات
Oman Refineries and Petrochemicals Company

AS ON DATE: 18TH MARCH 2010

Client : -

ORPC - Sohar Refinery, Sohar - Oman



الخدمات الفنية الخاصة ش.م.م.
SPECIAL TECHNICAL SERVICES L.L.C
(A MEMBER OF THE O.C.S.C. GROUP OF COMPANIES)

Contractor : -

STS - Special Technical Services L.L.C.

P.O. Box # 307, Al Harthy Complex, P.C. 118

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Contract # : -

41009221

Contract Title : -

Static Equipment Turnaround and Inspection Works

2010 Shutdown - Sohar Refinery

Shutdown Phase

Day of TA

43

Days Left

14

Cut off Date :- 20-Mar-10 @ 16:00 hrs

Distri. :-

STS (Muscat Office)

STS (Shutdown Site Office)

ORPC (Team Leaders + Head of Departments)

Prepared by :

STS - Shutdown Planning Department

Site Office, Sohar Refinery, Sohar

Rev. B



STS - Special Technical Services L.L.C.

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**Daily Progress Report
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery**

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Archives from Personal Files



Work Performed in Brief today

- 1) Blasting and painting of Z-6560 3A, V-5321, ME-5404-T04, T6531
- 2) Reinsulation in F-3301 and Reactor R-2001 (internal)
- 3) 928 major equipments have been attended and 869 boxed up to date
- 4) 1629 valves have been opened and 1567 boxed up to date
- 5) First steel plate for Sulfur Pit lowered after lifting the top RCC slab
- 6) Brickwork inside Furnaces F3651 and F3652 recommenced after arrival
- 7) Installation of trays inside C-1101 under progress after arrival
- 8) T-6531 under final inspection after cleaning and bottom painting on identified areas
- 9)
- 10)
- 11)
- 12)
- 13)
- 14)
- 15)
- 16)



Daily Progress Report
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery

EQUIPMENT REPORT

Sr.	Equipment Details	Size	Org. Plan	On Site
1	Mobile Telescopic Crane	20-30 Ton	8	4
2	Mobile Telescopic Crane	50 Ton	6	4
3	Mobile Telescopic Crane	75-90 Ton	2	1
4	Mobile Telescopic Crane	100 Ton	2	1
5	Mobile Telescopic Crane	160 Ton	1	1
6	Mobile Telescopic Crane	220 Ton	1	1
7	Mobile Telescopic Crane	300 Ton	1	0
8	Trailer & Bed,	40 Ton	2	6
9	Lorry, WITH HIAB	10 Ton	4	2
10	PLAIN TRUCK-Lorry Trk	10 T	4	2
11	Mobile Bucket Truck	30M reach	2	1
12	SODA Ash Washing skid Tank		0	0
13	Lorry Truck (Canter)	3T	9	5
14	Forklift,	10 Ton	0	
15	Forklift,	7 Ton	5	5
16	Forklift,	5 Ton	7	6
17	Forklift,	3 Ton	2	1
18	Welding Machine	400 amp	15	25
19	Mobile Air Compressor,	825 CFM	10	5
20	Mobile Air Compressor,	375 CFM	14	13
21	Generator Set,	50KV	12	8
22	Generator Set,	10KV	14	2
23	Generator Set,	100KVA	0	1
24	Generator Set,	500KVA	0	1
25	AC/DC Welding Machine		0	
26	Plasma Cutter		2	1
27	Diaphram Pump		3	
28	Hydrotest Pump c/w calib. Gauge		20	25
29	HP Water Jet Mach. –	20,000 psi	2	1
30	HP Water Jet Mach. –	15,000 psi	3	4
31	HP Water Jet Mach. –	10,000 psi	0	5
32	INRO-EXRO H.E. Bundle Auto washing machine		1	
33	H.E. Bundle Puller,	45 Ton	2	2
34	H.E. Bundle Puller,	40 Ton	2	1
35	FUEL TANKER		2	2
36	Water Tanker		2	3
37	VACUUM TRUCK	22kl	2	0
38	VACUUM TRUCK	5kl	2	2
39	SALOON CAR		4	4
40	D/C PICK UP		10	10
41	BUS	22 seat	6	8
42	BUS	52 seat	18	15
43	Bicycles		15	45
44	Tricycles		10	10
45	Vac. Unit for Catalyst handling		1	1
46	Sieving / Reclassifying Machine		1	1
47	Showel & winch		0	1
47	Sand / Grit Blasting Equipment		0	1
TOTAL			229	237



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*Daily Progress Report
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery*

AREAS OF CONCERN

1 Awaiting sleeves for E-1110 (10 banks)

2

3

4

5

6

7

8

9

10

Archives from Personal Files



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HSE STATISTICS

Static Equipment Turnaround and Inspection Works

2010 Shutdown - Sohar Refinery

STS COMPLETED 800,000 LTI FREE MANHOURS ON THE JOB ON 19TH MARCH

Area	AREA 1			AREA 2						AREA 3			AREA 4						AREA 5		
Unit	110	120	130	200	210	220	230	260	270	320	400	420	330	340	350	360	365	370	280	290	410
Name	CDU	CDU	CDU	RFCC	RFCC	RFCC	RFCC	RFCC	PRU	DUPU	TAME	GHDS	HGU	SWSU	ARU	SRU1	SRU2	TGTU	DHDS	IN-ALK	TAME
Hot Work Permit	3		1	3	2	2	1												1		
Cold Work Permit	5		1	3	6	3	1	2	1		1		1	3	1	2			1	1	3
Confined Space	2			2			1														
Heavy Lifts																					1
Radiography					1		1			1			1			1	1				
No. of Supervisors	6		1	6	4	4	3	2	1	1	1		2	2	1	2	1		2	1	2
Approx.No of Men	68		4	47	52	26	10	11	6	2	9		7	15	10	17	2		8	4	16

Area	AREA 6										AREA 7								OTHERS				
Unit	500	510	520	530	540	550	560	570	580	590	910	380	600	610	620	630	640	650	690	700	800	900	950
Name	UFC	PWR	SWI	WS	SCS	FG/NG	IA/PA	N2	CI	FO	FFS	SGU	OFF	FTK	PTK	TLF	MLF	WWT	FLR	MRN	BLD	COM	BATR
Hot Work Permit				2																			
Cold Work Permit				3								1											
Confined Space				4																			
Heavy Lifts																							
Radiography																							
No. of Supervisors				4								1											
Approx.No of Men				24								4											

Area	AREA 3								AREA 4								OTHERS				
Unit	320	400	420	330	340	350	360	365	370	380	600	610	620	630	640	650	690	700	800	900	950
Name	DUPU	TAME	GHDS	HGU	SWSU	ARU	SRU1	SRU2	TGTU	SGU	OFF	FTK	PTK	TLF	MLF	WWT	FLR	MRN	BLD	COM	BATR
Hot Work Permit																					
Cold Work Permit		1		1	3	1	2			1											
Confined Space																					
Heavy Lifts																					
Radiography	1			1			1	1													
No. of Supervisors	1	1		2	2	1	2	1													
Approx.No of Men	2	9		7	15	10	17	2													

MANPOWER:	INDIRECT	DIRECT	TOTAL	WORK HRS.	SPENT MHRS.	SPENT TO DATE
DAY SHIFT	215	1,014	1,229	12	11,368	714,426
NIGHT SHIFT	19	136	155	12	1,632	106,680
TOTAL	234	1,150	1,384	12	13,000	821,106
SINCE LAST LTI						821,106



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QA/QC STATISTICS
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery

RFI STATUS:

AREA	Today	Pre-Inspection	Preclean	Post Clean	Repairs	Re-Inspect	Hydro	Final Inspect	Boxup
AREA-1	6			1			2	1	2
AREA-2	16						10	2	4
AREA-3	2						1		1
AREA-4	11			2				5	4
AREA-5	2						2		
AREA-6	10			1				2	7
AREA-7	11							1	10
TOTAL	58	0	0	4	0	0	15	11	28

TQ STATUS:

AREA	RAISED	CLOSED
AREA-1	1	1
AREA-2	1	1
AREA-3	1	1
AREA-4	1	1
AREA-5		
AREA-6		
AREA-7	1	1
GENERAL	4	4

CUMULATIVE STATUS OF BOXUP CERTIFICATES

AREA	Yesterday	Today	Cumulative
AREA-1	146	3	149
AREA-2	139	4	143
AREA-3	126	1	127
AREA-4	130	6	136
AREA-5	92	0	92
AREA-6	82	7	89
AREA-7	43	10	53
TOTAL	758	31	789



<u>Shutdown Progress Summary</u> <u>Area Wise</u>					
Unit No. / Description	Wt %	Cumm. Planned %	Cumm. Actual %	Var %	Previous Day's Progress
Overall progress	100.00 %	98.88 %	99.13 %	0.25 %	0.24 %
AREA # 100 (CDU - Crude Distillation Unit)	11.71 %	99.06 %	99.11 %	0.05 %	0.00 %
AREA # 200 (RFCC - Residue Fluid Catalytic Reaction)	37.09 %	99.50 %	99.34 %	-0.16 %	0.00 %
AREA # 300 (GHDS - Gasoline HDS Units)	9.61 %	99.22 %	99.67 %	0.45 %	0.00 %
AREA # 400 (Sulphur Recovery / HG Units)	15.02 %	99.78 %	99.71 %	-0.07 %	0.00 %
AREA # 500 (GHDS / InAlk Units)	6.47 %	99.36 %	99.65 %	0.29 %	0.54 %
AREA # 600 (Utilities)	9.68 %	99.93 %	99.10 %	-0.83 %	0.00 %
AREA # 700 (Offsites)	6.97 %	###	99.17 %	-0.83 %	0.00 %
Rotating Equipment	3.45 %	87.50 %	91.84 %	4.34 %	5.84 %



Major Equipment Status - Stage Wise
Summary of Equipments (AREA-WISE)

Area No.	Unit No.	Equipment Type	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (tube side)	Reinstalled internals (manways / demisters / parts)	Pressure test (final test)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despaded	
Area # 100	Unit # 1100	Air Cooler - 49 banks	49	49	N/A	N/A	N/A	49	49	39	N/A	39	N/A	N/A	N/A	39	29	
		Column - 5 nos.	5	5	5	N/A	5	4	4	4	N/A	N/A	4	N/A	4	4	4	
		DesuperFurnace - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
		Exchanger - 32 nos.	32	32	N/A	32	N/A	32	32	30	32	30	N/A	29	N/A	30	15	
		Furnace - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	0	
		Strainer - 37 nos.	37	37	37	N/A	N/A	37	37	37	N/A	N/A	37	N/A	N/A	37	37	
		Vessel - 11 nos.	11	11	11	N/A	11	11	11	11	N/A	N/A	11	N/A	11	11	5	
	Unit # 1200	Absorber - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1	
		Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2	
		Strainer - 3 nos.	3	3	3	N/A	N/A	3	3	3	N/A	N/A	3	N/A	N/A	3	3	
		Vessel - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2	
	Unit # 1300	Absorber - 2 nos.	2	2	2	N/A	N/A	2	2	2	N/A	N/A	2	N/A	N/A	2	2	
		Strainer - 6 nos.	6	6	6	N/A	N/A	6	6	6	N/A	N/A	6	N/A	N/A	6	6	
		Vessel - 6 nos.	6	6	6	N/A	6	6	6	6	N/A	N/A	6	N/A	6	6	6	
	Unit # 6500	Exchanger - 3 nos.	3	3	N/A	3	N/A	3	3	3	3	3	N/A	3	N/A	3	3	
		Structural Works - 1 no.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	0	
		Vessel - 2 nos.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	0	
	Grand total for Area - 100			163	163	78	35	28	162	162	150	35	72	77	32	27	150	115
		Unit # 2000	Absorber - 5 nos.	5	5	5	N/A	N/A	5	5	4	N/A	N/A	4	N/A	N/A	4	0
			Exchanger - 7 nos.	8	8	N/A	8	N/A	8	8	8	8	8	N/A	8	N/A	8	0
Furnace - 2 nos.			2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	1	0	
Structural Works - 2 nos.			2	2	N/A	N/A	N/A	2	2	2	N/A	N/A	N/A	N/A	N/A	2	0	
Vessel - 4 nos.			4	4	4	N/A	4	4	4	4	N/A	N/A	4	N/A	4	2	0	
Whole Package - 1 no.			1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	0	
Unit # 2100		Air Cooler - 44 banks	44	44	N/A	N/A	N/A	44	44	44	N/A	44	N/A	N/A	N/A	44	6	
		Column - 3 nos.	3	3	3	N/A	3	2	2	2	N/A	N/A	2	N/A	2	2	0	
		Exchanger - 25 nos.	25	25	N/A	25	N/A	25	25	25	24	24	N/A	24	N/A	24	0	
		Strainer - 1 no.	0	0	0	N/A	N/A	0	0	0	N/A	N/A	0	N/A	N/A	0	0	

Area No.	Unit No.	Equipment Type	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (tube side)	Reinstalled internals (manways / demisters / parts)	Pressure test (final test)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despanded	
Area # 200		Vessel - 4 nos.	4	4	4	N/A	4	3	3	3	N/A	N/A	3	N/A	3	3	0	
	Unit # 2200	Air Cooler - 38 banks	38	38	N/A	N/A	N/A	38	38	38	N/A	38	N/A	N/A	N/A	38	27	
		Column - 5 nos.	5	5	5	N/A	5	5	5	5	N/A	N/A	5	N/A	5	5	1	
		Exchanger - 43 nos.	42	42	N/A	42	N/A	42	34	34	34	34	N/A	34	N/A	34	21	
		Ejector - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1	
		Strainer - 4 nos.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
		Vessel - 10 nos.	9	9	9	N/A	9	9	9	9	N/A	N/A	9	N/A	9	9	5	
		Whole Package - 1 no.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	1	
	Unit # 2300	Boiler - 3 no.	1	1	N/A	N/A	1	1	1	0	N/A	N/A	N/A	N/A	0	0	0	
		Exchanger - 6 nos.	6	6	N/A	6	N/A	6	6	6	6	6	N/A	5	N/A	6	2	
		Ejector - 6 nos.	5	5	5	N/A	N/A	5	5	5	N/A	N/A	5	N/A	N/A	5	5	
		Strainer - 3 nos.	3	3	3	N/A	N/A	2	1	1	N/A	N/A	1	N/A	N/A	0	0	
		Stack - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
		Structural Works - 7 nos.	7	7	N/A	N/A	N/A	7	7	7	N/A	N/A	N/A	N/A	N/A	7	7	
		Vessel - 8 nos.	8	8	8	N/A	8	8	7	7	N/A	N/A	6	N/A	6	5	0	
		Whole Package - 2 nos.	2	2	N/A	N/A	N/A	2	2	2	N/A	N/A	N/A	N/A	N/A	2	0	
	Unit # 2600	Absorber - 2 nos.	2	2	2	N/A	N/A	2	2	0	N/A	N/A	0	N/A	N/A	0	0	
		Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	0	
		Exchanger - 1 no.	1	1	N/A	1	N/A	1	1	1	1	1	N/A	1	N/A	1	0	
		Strainer - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
		Vessel - 10 nos.	10	10	10	N/A	10	10	10	10	N/A	N/A	8	N/A	8	7	0	
	Unit # 2700	Air Cooler - 1 bank	1	1	N/A	N/A	N/A	1	1	1	N/A	1	N/A	N/A	N/A	1	1	
		Column - 3 nos.	3	3	3	N/A	3	3	3	3	N/A	N/A	3	N/A	3	3	0	
		Exchanger - 23 nos.	23	23	N/A	23	N/A	23	23	23	23	23	N/A	23	N/A	23	0	
		Strainer - 4 nos.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
		Vessel - 10 nos.	8	8	8	N/A	8	8	8	8	N/A	N/A	7	N/A	7	6	0	
	Grand total for Area - 200			280	280	78	105	59	277	267	263	96	179	66	95	51	251	77
	0	Unit # 3200	Absorber - 3 nos.	3	3	3	N/A	N/A	3	3	3	N/A	N/A	3	N/A	N/A	3	3
			Air Cooler - 16 banks	16	16	N/A	N/A	N/A	16	16	16	N/A	16	N/A	N/A	N/A	16	14
			Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
			DesuperFurnace - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
			Exchanger - 26 nos.	26	26	N/A	26	N/A	26	26	24	26	26	N/A	26	N/A	26	25
Ejector - 1 no.			1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	0	
Furnace - 1 no.			1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1	
Strainer - 10 nos.			10	10	10	N/A	N/A	10	10	10	N/A	N/A	10	N/A	N/A	10	10	
Vessel - 14 nos.			14	14	14	N/A	14	14	14	14	N/A	N/A	14	N/A	14	14	14	

Area No.	Unit No.	Equipment Type	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (tube side)	Reinstalled internals (manways / demisters / parts)	Pressure test (final test)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despaded
Area # 300		Whole Package - 1 no.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	1
	Unit # 4000	Air Cooler - 6 banks	6	6	N/A	N/A	N/A	6	6	6	N/A	6	N/A	N/A	N/A	6	6
		Column - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
		Exchanger - 6 nos.	6	6	N/A	6	N/A	6	6	5	6	6	N/A	6	N/A	6	6
		Vessel - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
	Unit # 4200	Air Cooler - 6 banks	6	6	N/A	N/A	N/A	6	6	6	N/A	6	N/A	N/A	N/A	6	2
		Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
		Exchanger - 15 nos.	15	15	N/A	15	N/A	15	15	15	15	15	N/A	15	N/A	15	14
		Furnace - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
		Strainer - 2 nos.	2	2	2	N/A	N/A	2	2	2	N/A	N/A	2	N/A	N/A	2	2
		Vessel - 7 nos.	7	7	7	N/A	7	7	7	7	N/A	N/A	7	N/A	7	7	7
		Whole Package - 2 nos.	2	2	N/A	N/A	N/A	2	2	2	N/A	N/A	N/A	N/A	N/A	2	2
	Grand total for Area - 300			125	125	47	47	30	125	125	122	47	75	47	47	30	125
Area # 400	Unit # 3300	Air Cooler - 4 banks	4	4	N/A	N/A	N/A	4	4	4	N/A	4	N/A	N/A	N/A	4	4
		Column - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
		Exchanger - 12 nos.	12	12	N/A	12	N/A	12	12	12	12	12	N/A	12	N/A	12	12
		Furnace - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
		Strainer - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Stack - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Vessel - 17 nos.	17	17	17	N/A	17	17	17	17	N/A	N/A	17	N/A	17	17	17
		Whole Package - 3 nos.	3	3	N/A	N/A	N/A	3	3	3	N/A	N/A	N/A	N/A	N/A	3	0
	Unit # 3400	Air Cooler - 6 banks	6	6	N/A	N/A	N/A	6	6	6	N/A	6	N/A	N/A	N/A	6	6
		Column - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	0	0
		Exchanger - 6 nos.	6	6	N/A	6	N/A	6	6	6	6	6	N/A	6	N/A	6	6
		Strainer - 2 nos.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Vessel - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
	Unit # 3500	Air Cooler - 4 banks	4	4	N/A	N/A	N/A	4	4	4	N/A	4	N/A	N/A	N/A	4	4
		Column - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	0
		Exchanger - 7 nos.	7	7	N/A	7	N/A	7	7	6	7	6	N/A	6	N/A	6	6
		Vessel - 3 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
		Whole Package - 1 no.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	0
	Unit # 3600	Absorber - 2 nos.	2	2	2	N/A	N/A	2	2	2	N/A	N/A	2	N/A	N/A	2	0
		Exchanger - 9 nos.	9	9	N/A	9	N/A	9	9	9	9	9	N/A	9	N/A	6	3
		Furnace - 2 nos.	2	2	2	N/A	1	1	1	0	N/A	N/A	0	N/A	0	0	0
		Vessel - 6 nos.	6	6	6	N/A	6	6	6	6	N/A	N/A	6	N/A	6	6	6
			Exchanger - 8 nos.	8	8	N/A	8	N/A	8	8	7	8	7	N/A	4	N/A	7

Area No.	Unit No.	Equipment Type	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (tube side)	Reinstalled internals (manways / demisters / parts)	Pressure test (final test)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despanded
	Unit # 3650	Furnace - 2 nos.	2	2	2	N/A	0	2	2	0	N/A	N/A	0	N/A	0	0	0
		Vessel - 6 nos.	6	6	6	N/A	6	6	6	6	N/A	N/A	6	N/A	6	6	6
		Whole Package - 1 no.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	0
	Unit # 3700	Air Cooler - 2 banks	2	2	N/A	N/A	N/A	2	2	2	N/A	2	N/A	N/A	N/A	2	2
		Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
		Exchanger - 7 nos.	7	7	N/A	7	N/A	7	7	7	7	7	N/A	7	N/A	7	7
		Furnace - 4 nos.	4	4	4	N/A	4	4	4	4	N/A	N/A	4	N/A	4	4	4
		Strainer - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Stack - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Vessel - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
	Grand total for Area - 400		125	125	55	49	45	124	124	119	49	63	51	44	44	115	97
Area # 500	Unit # 2800	Air Cooler - 1 bank	1	1	N/A	N/A	N/A	1	1	1	N/A	1	N/A	N/A	N/A	1	1
		Column - 2 nos.	2	2	2	N/A	2	2	2	2	N/A	N/A	2	N/A	2	2	2
		Exchanger - 3 nos.	3	3	N/A	3	N/A	3	3	3	3	3	N/A	3	N/A	3	3
		Strainer - 1 no.	1	1	1	N/A	N/A	1	1	1	N/A	N/A	1	N/A	N/A	1	1
		Vessel - 4 nos.	4	4	4	N/A	4	4	4	4	N/A	N/A	4	N/A	4	4	4
	Unit # 2900	Air Cooler - 8 banks	8	8	N/A	N/A	N/A	8	8	8	N/A	8	N/A	N/A	N/A	8	6
		Column - 3 nos.	3	3	3	N/A	3	3	3	3	N/A	N/A	3	N/A	3	3	1
		Exchanger - 22 nos.	22	22	N/A	22	N/A	22	22	22	22	22	N/A	22	N/A	22	21
		Strainer - 3 nos.	3	3	3	N/A	N/A	3	3	3	N/A	N/A	3	N/A	N/A	3	3
		Vessel - 5 nos.	5	5	5	N/A	5	5	4	5	N/A	N/A	5	N/A	5	5	5
	Unit # 4100	Air Cooler - 19 banks	19	19	N/A	N/A	N/A	19	19	19	N/A	19	N/A	N/A	N/A	19	19
		Column - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
		Exchanger - 9 nos.	9	9	N/A	9	N/A	9	9	9	9	9	N/A	9	N/A	9	7
		Strainer - 5 nos.	5	5	5	N/A	N/A	5	5	5	N/A	N/A	5	N/A	N/A	5	5
		Vessel - 5 nos.	5	5	5	N/A	5	5	5	5	N/A	N/A	5	N/A	5	5	5
	Grand total for Area - 500		91	91	29	34	20	91	90	91	34	62	29	34	20	91	84
Area # 600	Unit # 5100	Air Cooler - 6 banks	6	6	N/A	N/A	N/A	6	6	6	N/A	6	N/A	N/A	N/A	6	6
		Boiler - 2 nos.	2	2	N/A	N/A	2	2	2	2	N/A	N/A	N/A	N/A	2	2	2
		Exchanger - 13 nos.	13	13	N/A	13	N/A	13	13	13	13	13	N/A	13	N/A	13	13
		Strainer - 4 nos.	4	4	4	N/A	N/A	4	4	4	N/A	N/A	4	N/A	N/A	4	4
		Vessel - 7 nos.	7	7	7	N/A	7	7	7	7	N/A	N/A	7	N/A	7	7	7
	Unit # 5200	Structural Works - 5 nos.	5	5	N/A	N/A	N/A	5	5	5	N/A	N/A	N/A	N/A	N/A	5	5
	Unit # 5300	Exchanger - 3 nos.	2	2	N/A	2	N/A	2	2	1	2	2	N/A	1	N/A	2	1
		Strainer - 3 nos.	3	3	3	N/A	N/A	3	3	3	N/A	N/A	3	N/A	N/A	3	3
		Vessel - 7 nos.	7	7	7	N/A	7	6	6	6	N/A	N/A	6	N/A	6	6	6

Area No.	Unit No.	Equipment Type	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (tube side)	Reinstalled internals (manways / demisters / parts)	Pressure test (final test)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despaded
Area # 600	Unit # 5400	Exchanger - 7 nos.	7	7	N/A	7	N/A	7	7	7	7	7	N/A	7	N/A	7	7
		Vessel - 10 nos.	10	10	10	N/A	10	10	10	10	N/A	N/A	10	N/A	10	10	10
	Unit # 5500	Exchanger - 2 nos.	2	2	N/A	2	N/A	2	2	2	2	2	N/A	2	N/A	2	2
		Vessel - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
	Unit # 5600	Exchanger - 12 nos.	12	12	N/A	12	N/A	12	12	12	12	12	N/A	12	N/A	12	12
		Vessel - 5 nos.	4	4	4	N/A	4	4	4	4	N/A	N/A	4	N/A	4	3	2
		Whole Package - 2 nos.	1	1	N/A	N/A	N/A	1	1	1	N/A	N/A	N/A	N/A	N/A	1	1
	Unit # 5800	Exchanger - 1 no.	1	1	N/A	1	N/A	1	1	1	1	1	N/A	1	N/A	1	1
	Grand total for Area -600			87	87	36	37	31	86	86	85	37	43	35	36	30	85
Area # 700	Unit # 3800	Exchanger - 1 no.	1	1	N/A	1	N/A	0	0	0	0	0	N/A	0	N/A	0	0
		Structural Works - 5 nos.	5	5	N/A	N/A	N/A	5	5	5	N/A	N/A	N/A	N/A	N/A	5	5
		Vessel - 3 nos.	3	3	3	N/A	3	3	3	2	N/A	N/A	2	N/A	2	2	2
	Unit # 6100	Exchanger - 1 no.	1	1	N/A	1	N/A	1	1	1	1	1	N/A	1	N/A	1	1
	Unit # 6300	Exchanger - 1 no.	1	1	N/A	1	N/A	1	1	1	1	1	N/A	1	N/A	1	1
		Vessel - 1 no.	1	1	1	N/A	1	1	1	1	N/A	N/A	1	N/A	1	1	1
	Unit # 6500	Structural Works - 11 nos.	11	11	N/A	N/A	N/A	11	11	11	N/A	N/A	N/A	N/A	N/A	11	0
		Whole Package - 1 no.	1	1	N/A	N/A	N/A	0	0	0	N/A	N/A	N/A	N/A	N/A	0	0
	Unit # 6900	Flare - 2 nos.	2	2	N/A	N/A	N/A	2	2	2	N/A	N/A	N/A	N/A	N/A	2	2
Vessel - 3 nos.		3	3	3	N/A	3	3	3	3	N/A	N/A	3	N/A	3	3	3	
Grand total for Area - 700			29	29	7	3	7	27	27	26	2	2	6	2	6	26	15
Grand total for Area - 700			57	57	14	5	14	54	54	52	4	4	12	4	12	52	30



STS - Special Technical Services L.L.C.

Other Equipment - Progress by Stage
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery

STS

Stage wise completion of Work(Summary by Equipment Type per Unit)																	
Eqpt	Area	Packages	Open	Clean	Inspect	Boxup											
Tanks	100	2 Tanks	2	2	2	2											
	200	5 Tanks	4	4	4	4											
	300	4 Tanks	4	4	4	4											
	400																
	500	1 Tanks	2	2	2	2											
	600	25 Tanks	24	24	24	24											
	700	2 Tanks	2	2	2	2											
	Others																
Eqpt	Area	Packages	Open	Access	Cleaning in Progress	Removal in Progress	Removal Complete	Cleaning Complete	Final Inspection								
Pits	400	2 Pits	1	1	1	1	1	1									
	500	4 Pits	4	4	4	4	4	4	4								
	600	7 Pits	6	5	5	5	5	4	4								
	700	26 Pits	25	25	25	25	20	18	18								
Eqpt	Area	Packages	QA ready	Scaffold	Blind	RFI/Permit	Prefab	Field Fab	Hydrotest	Inspect	Release	Deblind					
Piping & Hotworks	100	11 Packs	9	9	9	9	9	9	8	8	7	7					
	200	38 Packs	36	36	34	34	30	30	28	25	20	19					
	300	11 Packs	11	11	11	11	11	11	10	10	9	9					
	400	19 Packs	15	15	15	15	12	12	12	10	8	8					
	500	14 Packs	12	12	12	12	12	10	10	10	8	8					
	600	4 Packs	3	3	3	3	3	3	2	2	2	2					
	700	14 Packs	14	14	14	14	14	14	14	14	14	14					
	Others																
Eqpt	Area	Packages	Blind	Access	Remove	Furbish	Install										
Equipment	100																
	200	28 Packs	28	26	22	n/a	20										
	300	6 Packs	6	6	5	n/a	5										
	400	21 Packs	21	18	16	n/a	15										

Rotating Ec	500	2 Packs	2	2	2	n/a	2											
	600	43 Packs	40	40	33	n/a	30											
	700	43 Packs	40	40	36	n/a	34											
	Others																	

Archives from Personal Files



<u>Valves Stage Wise Progress</u> <u>Summary of Miscellaneous Equipment (AREA - WISE)</u>									
Area No.	Unit No.	Equipment Type	Opened / removed	Transported to W/S	Overhauled / repaired	Inspected	Transported to Site	Box-up / reinstalled	Remarks
Area # 100	Unit # 1100	Control Valves - 36 nos.	36	36	36	36	36	36	completed
		Flow Elements - 14 nos.	14	N/A	14	14	N/A	14	completed
		Miscellaneous Valves - 8 nos.	8	8	7	7	7	7	
		Relief Valves - 62 nos.	62	62	62	62	62	55	
	Unit # 1200	Control Valves - 4 nos.	4	4	4	4	4	4	completed
		Flow Elements - 1 no.	1	N/A	1	1	N/A	1	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 10 nos.	10	10	10	10	10	10	completed
	Unit # 1300	Control Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 25 nos.	25	25	25	25	25	25	completed
	Grand total for Area - 100		162	147	161	161	146	154	
Area # 200	Unit # 2000	Control Valves - 27 nos.	27	27	24	24	24	24	
		Miscellaneous Valves - 15 nos.	15	13	12	12	12	12	
		Relief Valves - 31 nos.	31	31	31	31	31	31	completed
		Slide Valves - 5 nos.	5	N/A	5	5	N/A	4	
	Unit # 2100	Control Valves - 17 nos.	17	17	11	11	11	9	
		Flow Elements - 16 nos.	16	N/A	16	16	N/A	16	completed
		Motor Operated Valves - 2 nos.	2	2	2	2	2	2	cancelled
		Miscellaneous Valves - 13 nos.	13	13	6	4	4	4	
		Relief Valves - 26 nos.	26	26	26	26	26	25	
	Unit # 2200	Control Valves - 17 nos.	17	17	17	17	17	17	completed
		Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
		Miscellaneous Valves - 3 nos.	3	3	3	3	3	3	completed
		Relief Valves - 44 nos.	44	44	43	43	43	43	
	Unit # 2300	Control Valves - 27 nos.	27	27	24	24	24	20	
		Flow Elements - 4 nos.	4	N/A	4	4	N/A	4	completed
		Miscellaneous Valves - 9 nos.	9	9	4	4	4	4	
		Relief Valves - 41 nos.	41	41	40	40	40	38	
		Slide Valves - 2 nos.	2	N/A	2	2	N/A	2	completed
	Unit # 2600	Control Valves - 15 nos.	15	15	15	15	15	15	completed
		Miscellaneous Valves - 2 nos.	2	2	2	2	2	2	completed
		Relief Valves - 23 nos.	23	23	23	23	23	23	completed
	Unit # 2700	Control Valves - 32 nos.	32	32	32	32	32	32	completed
		Flow Elements - 3 nos.	3	N/A	3	3	N/A	3	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 29 nos.	29	29	29	29	29	29	completed
	Grand total for Area - 200		406	372	377	375	343	365	
0	Unit # 3200	Control Valves - 25 nos.	25	25	24	24	24	24	
		Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
		Miscellaneous Valves - 7 nos.	7	7	7	7	7	7	completed

Area No.	Unit No.	Equipment Type	Opened / removed	Transported to W/S	Overhauled / repaired	Inspected	Transported to Site	Box-up / reinstalled	Remarks
Area # 300		Relief Valves - 88 nos.	88	88	88	88	88	88	completed
	Unit # 4000	Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
		Miscellaneous Valves - 3 nos.	3	3	3	3	3	3	completed
		Relief Valves - 6 nos.	6	6	6	6	6	6	completed
		Control Valves - 10 nos.	10	10	10	10	10	10	completed
	Unit # 4200	Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
	Relief Valves - 25 nos.	25	25	25	25	25	25	completed	
Grand total for Area - 300			170	164	169	169	163	169	
Area # 400	Unit # 3300	Control Valves - 90 nos.	90	90	90	90	90	90	completed
		Miscellaneous Valves - 7 nos.	7	7	7	7	7	7	completed
		Relief Valves - 59 nos.	59	59	59	59	59	59	completed
	Unit # 3400	Control Valves - 5 nos.	5	5	5	5	5	5	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	cancelled
		Relief Valves - 7 nos.	7	7	5	5	5	5	
	Unit # 3500	Control Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 12 nos.	12	12	12	12	12	12	completed
	Unit # 3600	Control Valves - 7 nos.	7	7	7	7	7	7	completed
		Flow Elements - 6 nos.	6	N/A	6	6	N/A	6	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 22 nos.	22	22	22	22	22	22	completed
	Unit # 3700	Control Valves - 7 nos.	7	7	7	7	7	7	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 18 nos.	18	18	18	18	18	18	completed
Grand total for Area - 400			244	238	242	242	236	242	
Area # 500	Unit # 2800	Flow Elements - 1 no.	1	N/A	1	1	N/A	1	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 17 nos.	17	17	17	17	17	17	completed
	Unit # 2900	Control Valves - 8 nos.	8	8	8	8	8	8	completed
		Flow Elements - 4 nos.	4	N/A	4	4	N/A	4	completed
		Miscellaneous Valves - 3 nos.	3	3	3	3	3	3	completed
		Relief Valves - 45 nos.	45	45	45	45	45	45	completed
	Unit # 4100	Control Valves - 4 nos.	4	4	4	4	4	2	
		Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 22 nos.	22	22	22	22	22	22	completed
Grand total for Area - 500			108	101	108	108	101	106	
Area # 600	Unit # 5100	Control Valves - 9 nos.	9	9	9	9	9	9	completed
		Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
		Miscellaneous Valves - 9 nos.	9	9	9	9	9	9	completed
		Relief Valves - 64 nos.	64	64	64	64	64	64	completed
	Unit # 5200	Control Valves - 5 nos.	5	5	5	5	5	5	completed
	Unit # 5300	Miscellaneous Valves - 3 nos.	3	3	3	3	3	3	completed
		Relief Valves - 40 nos.	40	40	40	40	40	40	completed
	Unit # 5400	Control Valves - 38 nos.	38	38	38	38	38	38	completed
		Flow Elements - 4 nos.	4	N/A	4	4	N/A	4	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 64 nos.	64	64	64	64	64	64	completed
	Unit # 5500	Control Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 4 nos.	4	4	4	4	4	4	completed
	Unit # 5600	Control Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 33 nos.	33	33	33	33	33	33	completed
	Unit # 5800	Relief Valves - 1 no.	1	1	1	1	1	1	completed

Area No.	Unit No.	Equipment Type	Opened / removed	Transported to W/S	Overhauled / repaired	Inspected	Transported to Site	Box-up / reinstalled	Remarks
	Unit # 9100	Flow Elements - 1 no.	1	N/A	1	1	N/A	1	completed
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 12 nos.	12	12	8	8	8	7	
		Grand total for Area - 600	293	286	289	289	282	288	
Area # 700	Unit # 3800	Control Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 1 no.	1	1	1	1	1	1	completed
		Slide Valves - 1 no.	1	N/A	1	1	N/A	1	completed
	Unit # 6100	Control Valves - 4 nos.	4	4	4	4	4	4	completed
		Motor Operated Valves - 1 no.	1	1	1	1	1	1	cancelled
		Miscellaneous Valves - 1 no.	1	1	1	1	1	1	completed
		Relief Valves - 27 nos.	27	27	27	27	27	27	completed
	Unit # 6200	Control Valves - 11 nos.	11	11	11	11	11	11	completed
		Relief Valves - 134 nos.	134	134	134	134	134	134	completed
	Unit # 6300	Relief Valves - 8 nos.	8	8	8	8	8	8	completed
	Unit # 6400	Motor Operated Valves - 9 nos.	9	9	9	9	9	9	cancelled
	Unit # 6500	Control Valves - 2 nos.	2	2	2	2	2	2	completed
		Flow Elements - 1 no.	1	N/A	1	1	N/A	1	completed
		Relief Valves - 43 nos.	43	43	42	42	42	40	
	Unit # 6900	Flow Elements - 2 nos.	2	N/A	2	2	N/A	2	completed
	Grand total for Area - 700		246	242	245	245	241	243	



STS - Special Technical Services L.L.C.

EQUIPMENT WISE- TOTAL SCOPE SUMMARY
Static Equipment Turnaround and Inspection Works
2010 Shutdown - Sohar Refinery

STS

Major Equipment Status
Overall Summary of Equipments

Equipment Type	Total as per scope	Spaded	Opened (plugs / covers / manholes)	Removed internals (manways / demisters)	Pull bundle	Erected internal scaffolding	Cleaned internally (equipment / bundle)	Inspected	Repaired if any	Reinstalled (bundle)	Pressure test (shell side)	Reinstalled internals (manways / demisters / parts)	Pressure test (tube side)	Dismantled internal scaffolding	Boxed up (plugs, covers / manholes)	Despaded
Absorber	15 nos.	15	15	15	N/A	N/A	15	15	12	N/A	N/A	12	N/A	N/A	12	6
Air Coolers	210 nos.	210	210	N/A	N/A	N/A	210	210	200	N/A	N/A	N/A	200	N/A	200	133
Boiler	3 nos.	3	3	N/A	N/A	3	3	3	2	N/A	N/A	N/A	N/A	2	2	2
Column	36 nos.	36	36	36	N/A	36	34	34	34	N/A	N/A	34	N/A	34	33	19
De-Superheater	2 nos.	2	2	2	N/A	N/A	2	2	2	N/A	N/A	2	N/A	N/A	2	1
Ejector	8 nos.	7	7	7	N/A	N/A	7	7	7	N/A	N/A	7	N/A	N/A	7	6
Exchanger	311 nos.	310	310	N/A	310	N/A	309	301	293	300	296	293	296	N/A	290	189
Flare	2 nos.	2	2	N/A	N/A	N/A	2	2	2	N/A	N/A	N/A	N/A	N/A	2	2
Furnace / Heaters	14 nos.	14	14	14	N/A	11	13	13	10	N/A	N/A	10	N/A	10	9	7
Stack	3 nos.	3	3	3	N/A	N/A	3	3	3	N/A	N/A	3	N/A	N/A	3	2
Strainer	88 nos.	83	83	83	N/A	N/A	82	81	81	N/A	N/A	81	N/A	N/A	80	77
Structural	31 nos.	31	31	N/A	N/A	N/A	31	31	31	N/A	N/A	N/A	N/A	N/A	31	17
Vessel	172 nos.	170	170	170	N/A	170	168	166	166	N/A	N/A	162	N/A	162	156	121
Whole Package	15 nos.	14	14	N/A	N/A	N/A	13	13	13	N/A	N/A	N/A	N/A	N/A	13	5
Grand Total	910 nos.	900	900	330	310	220	892	881	856	300	296	604	496	208	840	587

Equipment Type	Total as per scope		Opened / removed	Transported to W/S	Overhauled / repaired	Inspected	Transported to Site	Box-up / reinstalled	Remarks
Control Valves	409	nos.	409	409	396	396	396	386	
Flow Elements	69	nos.	69	N/A	69	69	N/A	69	Completed
Miscellaneous Valves	92	nos.	78	74	58	56	56	54	
Motor Operated Valve	10	nos.	0	0	0	0	0	0	Cancelled
Relief Valves	1043	nos.	1010	1010	1000	1000	1000	988	
Slide Valves	8	nos.	8	N/A	8	8	N/A	7	
Total			1574	1493	1531	1529	1452	1504	

Archives from Personal Files

	6-Feb	7-Feb	8-Feb	9-Feb	10-Feb	11-Feb	12-Feb	13-Feb	14-Feb	15-Feb	16-Feb	17-Feb	18-Feb	19-Feb	20-Feb	21-Feb	22-Feb	23-Feb	24-Feb	25-Feb	26-Feb	27-Feb	28-Feb	1-Mar	2-Mar	3-Mar	4-Mar	5-Mar	6-Mar	7-Mar	8-Mar	9-Mar	10-Mar	11-Mar	12-Mar	13-Mar	14-Mar	15-Mar	16-Mar	17-Mar	18-Mar	19-Mar	20-Mar	21-Mar	22-Mar	23-Mar	24-Mar	25-Mar	26-Mar	27-Mar	28-Mar	29-Mar	30-Mar	31-Mar	1-Apr	2-Apr																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Cumulative Project Planned Progress %	0.00%	0.01%	0.09%	0.20%	0.38%	1.23%	2.87%	5.13%	7.93%	11.51%	14.14%	17.11%	20.59%	24.06%	27.39%	30.97%	34.11%	37.93%	41.48%	44.65%	47.93%	50.56%	53.84%	57.39%	63.26%	67.25%	71.25%	74.31%	78.41%	82.27%	86.48%	89.32%	91.64%	93.28%	95.12%	95.83%	96.51%	96.99%	97.57%	97.96%	98.52%	98.88%	99.12%	99.27%	99.41%	99.51%	99.61%	99.70%	99.74%	99.81%	99.82%	99.90%	99.96%	99.96%	100.00%	100.00%																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Cumulative Spent Manhours	12916	28876	45832	63764	83660	103880	118532	136136	154760	173636	193136	212984	233072	253400	274160	295016	313196	334004	354752	375524	396260	417032	437648	458276	478976	499940	520892	541868	562880	584084	605156	626024	646844	667820	688056	707956	727760	746592	764212	780884	796428	808106	821106																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
Cumulative Earned Manhours	0	410	1642	2805	5815	11076	22037	39067	52817	68331	88299	113684	149052	181078	213018	243572	272223	304682	335323	358953	393316	420495	451050	483596	522633	555092	582790	607459	645804	665366	697652	719897	742142	765080	786027	804377	814504	835624	839260	844453	852849	855965	858043																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											

